Urshita Pal

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EDUCATION

University of Michigan, USA
Ph.D., Mathematics, Expected 2027.
Advisor: Jenny Wilson

University of Michigan, USA
M.S., Mathematics, 2021-23.

Chennai Mathematical Institute, India
B.Sc., Mathematics & Computer Science, 2018-21.

RESEARCH INTERESTS

(Co)homology of Arithmetic Groups, (Co)homological stability phenomena

AWARDS AND FELLOWSHIPS

Edwin Wilkinson Miller award
Summer 2026

Rackham Predoctoral Fellowship
Fall 2025 - Summer 2026

Rackham One-Term Dissertation Fellowship
January 2025 - April 2025

Gold Medal of Excellence, BSc Math & Computer Science
Chennai Mathematical Institute, 2021
Highest GPA at time of graduation

Tuition Fee Waiver
Chennai Mathematical Institute, Aug 2018 - April 2021

Bronze Medal, European Girls Math Olympiad
Held in Florence, Italy, 2018

International Math Olympiad Training Camp, Mumbai, India
Selected for Participation, 2017 & 2018

RESEARCH PAPERS

The Quillen and sharbly Hopf algebra structures on Steinberg homology coincide
with S. Payne
Arxiv preprint Arxiv:2609.38735

A projective resolution of the symplectic Steinberg module
Arxiv preprint Arxiv:2605.06499

The top cohomology of principal congruence subgroups of special linear groups over Euclidean number rings
Arxiv preprint Arxiv:2605.05087

Representation stability in the (co)homology of vertical configuration spaces
with D. Baron, C. Wang, J. Wilson, and C. Yang
ArXiv preprint ArXiv:2412.01128

TEACHING

University of Michigan
Primary Instructor, Math 105 (Data, Graphs, & Functions), Fall 2026
Primary Instructor, Math 105 (Data, Graphs, & Functions), Fall 2024
Primary Instructor, Math 115 (Calculus 1), Fall 2023
Primary Instructor, Math 115 (Calculus 1), Winter 2023
Primary Instructor, Math 115 (Calculus 1), Fall 2022
Primary Instructor, Math 115 (Calculus 1), Winter 2022
Primary Instructor, Math 115 (Calculus 1), Fall 2021

Chennai Mathematical Institute
Teaching Assistant, NPTEL Rings & Fields, Jan-March 2021
Teaching Assistant, Topology, Spring 2021
Teaching Assistant, Complex Analysis, Spring 2021
Teaching Assistant, Analysis 3, Fall 2020
Teaching Assistant, Probability Theory, Spring 2020

SERVICE AND MENTORSHIP

University of Michigan REU, Summer 2024
-*Graduate Student Mentor for the REU hosted by Jenny Wilson*

Lab of Geometry at Michigan
-*Served on the Admissions Committee in Fall '23 & Winter '24*

Student Dynamics/Geometry/Topology Seminar
-*Co-organiser 2022-23, Organiser 2023-24*

AWM Mentoring Programs
-*Mentor in the Mentor-Mentee program for the Academic Year 2022-23*
-*Participated in the 'Vertical Mentoring Program' in Winter 2024*

Michigan Directed Reading Program
-*Mentor in Winter 2023, Winter 2025, and Fall 2025*

Michigan Math Club
-*Fagnano's Problem and Reflecting Triangles*, Jan 2023

Michigan Math Circle
-*Tiling With Dominoes*, Feb 2022

TALKS

Invited Talks

-*The top cohomology of principal congruence subgroups of SL_nR and $Sp_{2n}R$*
Binghamton University, April 2026

-*The top cohomology of principal congruence subgroups of SL_nR and $Sp_{2n}R$*
Spring Redbud Topology Conference, University of Oklahoma, April 2026

-*Representation stability in the (co)homology of vertical configuration spaces*
Special Session at AMS Spring Sectional, Boston College, March 2026

-*The generalized Lee–Szczarba conjecture on the cohomology of principal congruence subgroups*
UC San Diego, February 2026

-*A Resolution of the Symplectic Steinberg Module*
Purdue University, October 2025

-*Steinberg Modules and Hopf Algebras* (expository)
University of Pennsylvania, March 2025

-*Steinberg Modules and $H^*(SL_n\mathbb{Z};\mathbb{Q})$* (expository)
Scissors Congruences, Algebraic K-Theory and Steinberg Modules, July 2024
(Held at the American Institute of Mathematics, Pasedena)

Contributed Talks

-*The top cohomology of congruence subgroups of SL_nR and $Sp_{2n}R$*
Algebraic Structures in Topology, San Juan, Puerto Rico, July 2026

-*Connectivity and Cohomology* (5 min lightning talk)
Young Geometric Group Theory, Bristol, April 2024

Summer Workshops

-*Segal's proof of homological stability* (expository)
Configuration Spaces Summer School, University of Michigan, May 2026

-*High Connectivity of Arc Complexes* (expository)
Talbot Workshop on Homological Stability, May 2025

CONFERENCES ATTENDED

- Algebraic Structures in Topology, San Juan, Puerto Rico, July 2026

- Configuration Spaces Summer School, University of Michigan, May 2026

- Spring Redbud Topology Conference, University of Oklahoma, April 2026

- Perspectives on the Cohomology of $\mathrm{GL}_n\mathbb{Z}$, Muenster, Germany, October 2025

- Scissors Congruences and Algebraic K-Theory, UPenn, July 2025

- Discrete Groups in Topology & Algebraic Geometry, Notre Dame, June 2025

- Polylogarithms, Homology of linear groups, & Steinberg Modules, June 2025
Held at the SwissMAP Research Station, Les Diablarets, Switzerland

- Talbot Workshop on Homological Stability, Cassapolis MI, May 2025

- Scissors Congruences, Algebraic K-Theory and Steinberg Modules, July 2024
Organised through the American Institute of Mathematics, Pasedena

- Young Geometric Group Theory XII, Bristol, April 2024

- Stability in Topology, Arithmetic and Representation Theory, Purdue, 2023
(Attended Virtually)

- Nearly Carbon Neutral Geometric Topology Conference, June 2023
(Attended Topic Group: Profinite and Residual Methods in Geometric Group Theory)

- Nearly Carbon Neutral Geometric Topology Conference, September 2022
(Attended Topic Group: Group Actions on Hyperbolic Spaces)